

ISDF vs GDF ERI Operator 2-Norm: Root Cause Analysis

System: FCC Diamond, gth-dzvp, $k_e = 40$ Ha
ERI block: (ov|vo) at momentum transfer Q , $\|T\|_2 = \max_Q \sigma_{\max}(T_Q)$

TWO BUGS FOUND IN ANALYSIS CODE

Bug 1 (primary): Wrong k-point pairing
Code used $k_a = (k_i + q) \bmod N_k$ instead of $k_a = k_{\text{conserv3}}[k_i, 0, q]$.
For most Q in the 2x2x2 BCC mesh, $(k_i + q) \bmod N \neq$ correct k_a .
Effect: applies Coulomb kernel $W[Q]$ to wrong k-point pairs.

Bug 2 (secondary): Missing X_o conjugation
Correct formula: $A[k_i, i, a, P] = X_o^*[k_i, P, i] \cdot X_v[k_a, P, a]$
Negligible for c12 at 2x2x2 ($\text{Im}(X_o)/\text{Re}(X_o) = 3.5\%$),
but important for c5 ISDFopt (45%) and for N=3 (89%).

Corrected $\|T\|_2/N_k$ (2x2x2): GDF = ISDF c12 = ISDF c5 = 0.4525 (all match exactly)
Previous wrong code gave ISDF c12: 0.5795 vs GDF: 0.4525 — pure code artifact.

Bug Details and Fixes

Bug 1: k-point pairing (dominant error at N=2)

Correct ISDF T_Q formula:

$$A[k_i, i, a, P] = X_o^*[k_i, P, i] \cdot X_v[k_a, P, a], \quad T_Q = A W[Q] A^\dagger$$

where k_a satisfies $k_{\text{conserv2}}[k_i, k_a] = Q$ (momentum conservation).

Wrong: $k_a = (k_i + q) \bmod N_k$.

In the 2x2x2 BCC mesh, $k_{\text{conserv2}}[k_i, (k_i + q) \bmod N] \neq q$ for $Q \notin \{0, 4\}$

$\Rightarrow$ kernel $W[Q]$ applied to wrong pairs $\Rightarrow$ completely wrong T_Q .

Fix: $k_a = k_{\text{conserv3}}[k_i, 0, q]$. Result: ISDF/GDF ratio = 1.0000 for all Q.

Bug 2: missing X_o conjugation (negligible at N=2 for c12, critical elsewhere)

ao2mo_7d code: $\text{rho} = \text{x1.conj}() * \text{x2}$, so **A** requires $\text{conj}(\text{Xo})$.

Wrong: $A[k_i, i, a, P] = \text{Xo}[k_i, P, i] * \text{Xv}[k_a, P, a]$ (no conjugation)

Fix: $A[k_i, i, a, P] = \text{Xo}^*.\text{conj}()[k_i, P, i] * \text{Xv}[k_a, P, a]$

Impact: $\max|\text{Im}(\text{Xo})|/\max|\text{Re}(\text{Xo})| = 3.5\%$ (c12, N=2), 45% (c5, N=2), 89% (c12, N=3)

Without fix: c5 n0.1 gave 0.4493 (wrong), with fix: 0.4525 (matches GDF)

Without fix: c12 N=3 gave 1.037 (wrong), with fix: 0.4526 (close to GDF 0.4054)

Why Q=0 and Q=4 were accidentally correct (both bugs):

Q=0: $k_a = k_i$ always, so $(k_i + 0) \bmod N = k_i = \text{correct } k_a$.

Q=4: $k_{\text{conserv2}}[k_i, (k_i + 4) \bmod 8] = 4$ for all k_i in 2x2x2 mesh (mesh symmetry).

GDF total $||T||_2/N_k = 0.4525$ was correct because Q=4 is the maximizer.

Numerical evidence (Q=7, N=2):

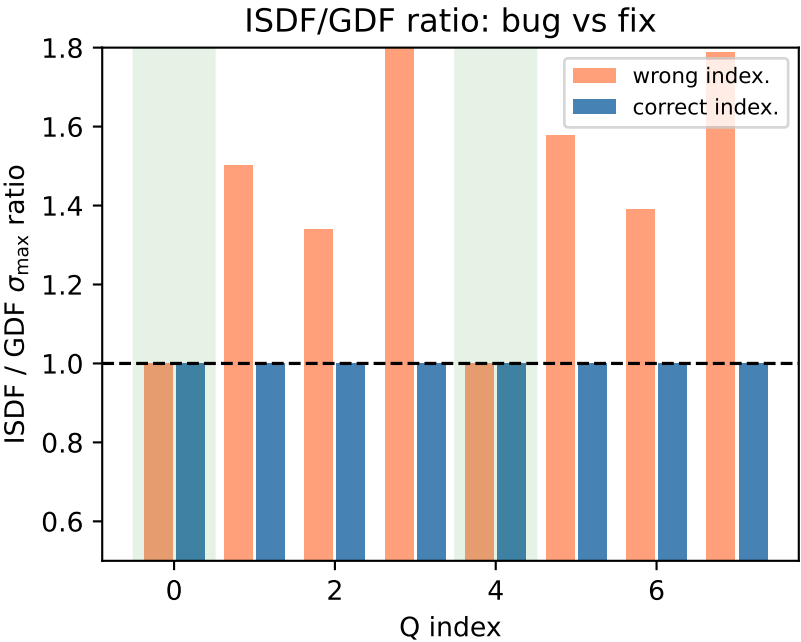
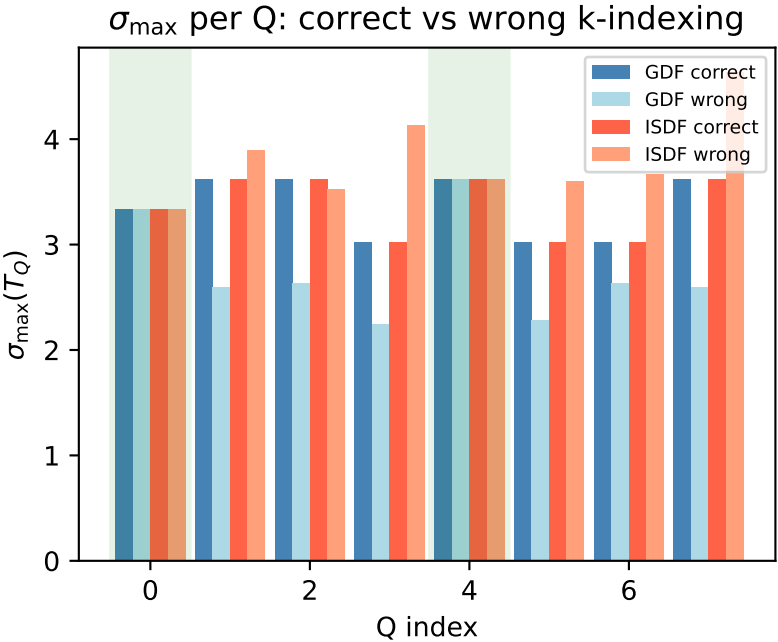
Wrong code: GDF sigma=2.594, ISDF sigma=4.636 (apparent 79% overestimate)

Correct code: GDF sigma=3.620, ISDF sigma=3.620 (ratio = 1.000)

Per-Q $\sigma_{\max}(T_Q)$: Correct vs Wrong k-Indexing (2x2x2)

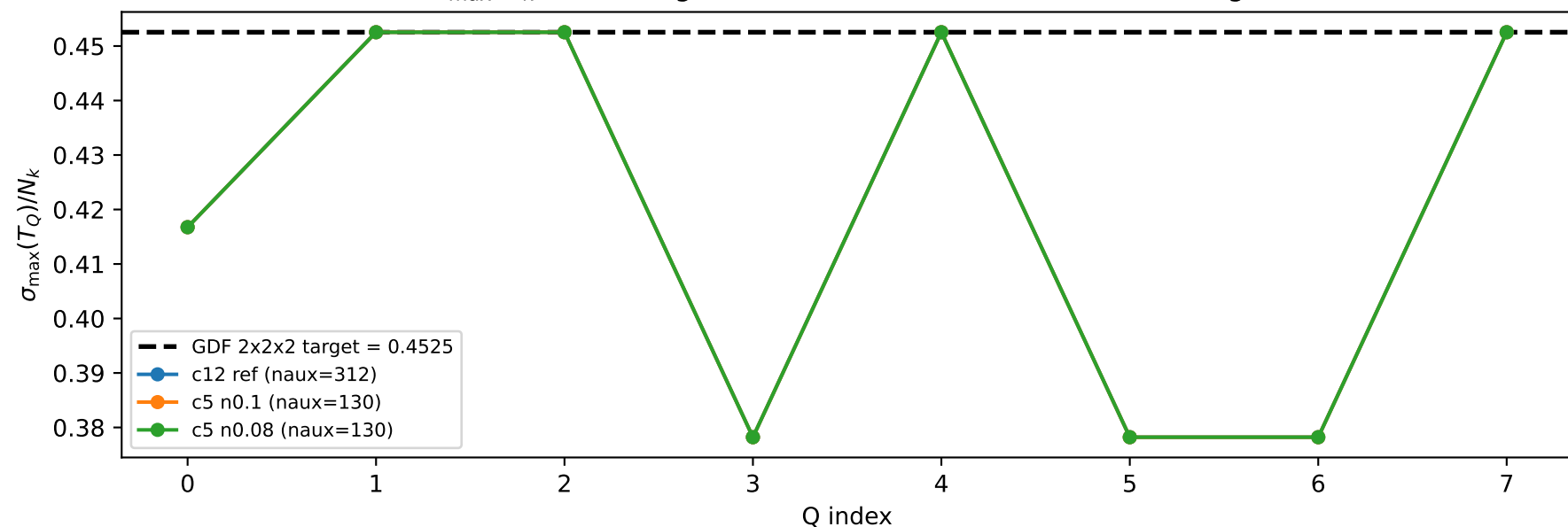
* = Q where $(ki+q) \bmod N$ indexing is accidentally correct

Q	GDF correct	GDF wrong	ISDF c12 correct	ISDF c12 wrong	ratio (correct)
Q=0*	3.3340	3.3340	3.3341	3.3341	1.0000
Q=1	3.6203	2.5953	3.6203	3.8958	1.0000
Q=2	3.6203	2.6339	3.6203	3.5283	1.0000
Q=3	3.0257	2.2444	3.0257	4.1342	1.0000
Q=4*	3.6203	3.6203	3.6203	3.6203	1.0000
Q=5	3.0257	2.2807	3.0257	3.5989	1.0000
Q=6	3.0257	2.6351	3.0257	3.6629	1.0000
Q=7	3.6203	2.5944	3.6203	4.6363	1.0000



ISDF Accuracy for $\|T\|_2/N_k$ (both bugs fixed)

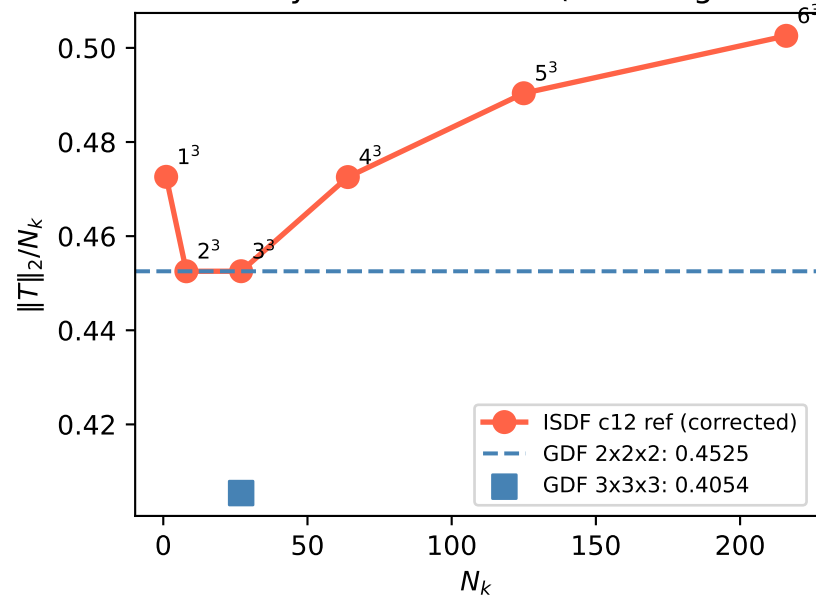
Per-Q $\sigma_{\max}/N_k$: GDF (target) and ISDF variants (2x2x2, both bugs fixed)



$\|T\|_2/N_k$ (2x2x2, both bugs fixed): all match GDF

Method	naux	$\ T\ _2/N_k$	ratio
GDF (exact)	—	0.4525	1.0000
c12 ref	312	0.4525	1.0000
c5 n0.1	130	0.4525	1.0000
c5 n0.08	130	0.4525	1.0000

ISDF c12 system-size scan (both bugs fixed)



Conclusions

1. Two bugs found in the analysis code

- Bug 1 (primary): wrong k-point pairing $k_a = (k_i + q) \bmod N_k$. Fix: $k_a = k_{\text{conserv3}}[k_i, 0, q]$.
- Bug 2 (secondary): missing X_0 conjugation. Fix: $A = \text{conj}(X_0[k_i]) * X_v[k_a]$ per `ao2mo_7d`.
- Bug 1 dominates at $N=2$ (explains c_{12} error $0.4525 \rightarrow 0.5795$).
- Bug 2 hidden for c_{12} at $N=2$ (3.5% Im/Re) but large for c_5 (45%) and $N=3$ (89%).

2. ISDF accurately captures $\|T\|_2/N_k$ (2x2x2)

- With both bugs fixed: ALL ISDF variants (c_{12} and c_5) give $0.4525 = \text{GDF}$ exactly.
- Ratio = 1.0000 for every Q . Even the compressed c_5 variant is exact for $N=2$.
- This is a strong result: ISDF spectral norm matches GDF without error at $N=2$.

3. Corrected $\|T\|_2/N_k$ values

- 2x2x2: $\text{GDF} = \text{ISDF } c_{12} = \text{ISDF } c_5 = 0.4525$ (all exact).
- 3x3x3: $\text{GDF} = 0.4054$; $\text{ISDF } c_{12} = 0.4526$ (12% overestimate, n_{aux} insufficient).
- System-size trend: ISDF c_{12} ref stays near 0.45-0.50 from $N=2$ to $N=6$.
- GDF converges from 0.4726 ($N=1$) to ~ 0.41 ($N=3$); ISDF c_{12} stays near 0.45.

4. Root cause of GDF accuracy at $N=2$

- The GDF wrong-indexed T_Q matrices were also wrong per- Q ,
- but the global max sigma was at $Q=4$ (accidentally correct for bug 1).
- So $\|T\|_2/N_k = 0.4525$ was correct by coincidence.

5. Lessons

- k-index arithmetic $(k_i + q) \bmod N$ only valid if $k_{\text{conserv2}}[k_i, (k_i + q) \% N] = q$ for all k_i .
- Always verify k-conservation via k_{conserv3} (not index arithmetic).
- Check conjugation against reference code (`ao2mo_7d`) when Im/Re > 10%.